

Tucker order-3 vs order-4

node207-23, H100 80GB, BF16 autocast, Llama geometry, seq 1024
16,384 tokens per optimizer step

1 Parameter count

Budget	Layout	Parameters	vs dense
133M	Tucker-4	132,998,816	-124,190,048
133M	Tucker-3: split input	133,002,112	-124,186,752
133M	Tucker-3: split output	132,999,936	-124,188,928
133M	Tucker-3: paired balanced	132,999,680	-124,189,184
225M	Tucker-4	225,023,872	-32,164,992
225M	Tucker-3: split input	225,000,704	-32,188,160
225M	Tucker-3: split output	224,997,120	-32,191,744
225M	Tucker-3: paired balanced	224,999,552	-32,189,312

2 Forward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	603.9	49.4
133M	Tucker-3: split input	600.5	102.7
133M	Tucker-3: split output	614.1	91.8
133M	Tucker-3: paired balanced	635.2	54.9
225M	Tucker-4	606.9	58.2
225M	Tucker-3: split input	646.3	110.2
225M	Tucker-3: split output	635.7	101.0
225M	Tucker-3: paired balanced	629.0	55.4

3 Backward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	731.6	87.8
133M	Tucker-3: split input	692.1	148.9
133M	Tucker-3: split output	729.3	173.2
133M	Tucker-3: paired balanced	962.6	83.0
225M	Tucker-4	741.7	113.0
225M	Tucker-3: split input	753.6	173.9
225M	Tucker-3: split output	751.3	203.8
225M	Tucker-3: paired balanced	965.5	88.2

4 Optimizer, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	83.7	67.3
133M	Tucker-3: split input	80.3	68.6
133M	Tucker-3: split output	80.5	61.5
133M	Tucker-3: paired balanced	76.8	58.8
225M	Tucker-4	88.4	73.3
225M	Tucker-3: split input	84.9	70.3
225M	Tucker-3: split output	96.2	84.4
225M	Tucker-3: paired balanced	78.8	65.5

5 Retraction, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	20.9	21.0
133M	Tucker-3: split input	95.5	96.9
133M	Tucker-3: split output	109.9	111.2
133M	Tucker-3: paired balanced	50.2	51.0
225M	Tucker-4	28.1	28.7
225M	Tucker-3: split input	384.4	384.3
225M	Tucker-3: split output	365.2	365.3
225M	Tucker-3: paired balanced	102.9	103.9

6 Full step, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	1445.2	226.1
133M	Tucker-3: split input	1473.5	418.0
133M	Tucker-3: split output	1536.3	438.7
133M	Tucker-3: paired balanced	1728.9	248.8
225M	Tucker-4	1468.7	275.1
225M	Tucker-3: split input	1872.7	739.9
225M	Tucker-3: split output	1854.4	756.4
225M	Tucker-3: paired balanced	1780.4	314.6

7 Forward+backward peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
133M	Tucker-3: paired balanced	3.08	8.95
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	4.19	9.72
225M	Tucker-3: split output	4.20	9.70
225M	Tucker-3: paired balanced	4.21	9.70

8 Full-step peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
133M	Tucker-3: paired balanced	3.08	8.95
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	5.60	9.72
225M	Tucker-3: split output	4.51	9.70
225M	Tucker-3: paired balanced	4.21	9.70

9 Fastest order-3 full step

Budget	MBxAcc	Layout	ms	time/Tucker-4
133M	1x16	Tucker-3: split input	1473.5	1.020x
133M	16x1	Tucker-3: paired balanced	248.8	1.101x
225M	1x16	Tucker-3: paired balanced	1780.4	1.212x
225M	16x1	Tucker-3: paired balanced	314.6	1.144x

10 Run

Field	Value
Commit	b9d90d5
Run	node207-23/gpu6
GPU	NVIDIA H100 80GB HBM3
PyTorch	2.9.1+cu128
CUDA	12.8
Samples	3 warmup + 12 measured

Every order-3 layout has three trainable factors and one fixed singleton buffer. Paired balanced groups three exact input/output factor pairs; split-input/output leave one side unsplit. Ratios above 1 mean slower than Tucker-4.